

Alejandro Jaimes

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science; Minor in Robotics

Orlando, FL

Expected Grad: May 2028

EXPERIENCE

Software Engineer Intern

BNY

Jul 2026 - Present

Lake Mary, FL

- Architected scalable cloud infrastructure pipeline supporting **3** microservices across **1** environment.
- Reduced system downtime by **20%** via real-time **Prometheus** and **Grafana** monitoring across **3** services.
- Engineered containerized **Docker** deployment workflow cutting provisioning time by **35%** for **4+** engineers.
- Designed **PostgreSQL** schema handling **1K+** daily records with **4ms** average query latency across **2** services.

Software Engineer Intern

Institute for Simulation and Training

May 2026 - Present

Orlando, FL

- Deployed spatial-anchor system with **10+** concurrent **XR** users, validated by **300+** participants over **15** hours.
- Streamed anchor-relative transforms via **Photon RPC** at **10Hz**, enabling real-time co-located multi-user tracks.
- Constructed re-localization system to correct drift every **2s**, sustaining accuracy of user movement across **50m**.
- Built **Firebase RTDB** anchor sync pipeline, eliminating manual setup, saving an estimated **20+ hours**.

President

Graphics Programming Knights

Apr 2026 - Present

Orlando, FL

- Founded Graphics Programming Knights, a non-profit student organization, growing membership to **200+**.
- Conducted **5+ workshops** on **GPU programming**, **mathematics**, and **game engines** with **70+** attendees.
- Led a team of **9+** engineers for the organization's **open-source** monorepository serving our **200+** members.
- Expanded club reach by **40%**, establishing partnerships to have workshops and hackathons with **3** organizations.

PROJECTS

HIP Renderer | C++, HIP/CUDA, ROCm, Python, Linux

- Built a **GPU-accelerated** path tracer in **C++/HIP** achieving **8.8x** speedup over CPU on **AMD MI300X**.
- Profiled kernel performance with **rocprof**, improving bandwidth by **55%** via **shared memory optimization**.
- Automated **GPU profiling** pipeline in **Python** analyzing **5** hardware metrics per optimization cycle on **Linux**.

greghouse | C++, Arduino, ESP32, React, AWS, Node.js, Lambda, DynamoDB

- **Placed 3rd** in the **Best Overall** category at **BloomKnights** out of **86** projects and **240+** participants.
- Implemented **ESP32** firmware in **C++** streaming soil telemetry to **AWS** every **2s** with **sub-1s latency**.
- Designed IoT pipeline with **API Gateway**, **Lambda**, and **DynamoDB** handling **1,800+** readings **hourly**.

sFun (SNES Emulator) | C++, CMake, JSON, SDL2

- Architected a **65C816 CPU** emulator in **C++** with 8/16-bit width switching, and 24-bit addressing.
- Built a **256-entry function pointer** dispatch table decomposed in **9** logical translation units for maintainability.
- Engineered a **JSON-driven harness** with register-level diffing, achieving **100%** pass rate across **25+** test cases.

Next Step | React, Tailwind, Next.js, TypeScript, Python, Flask, Vapi

- **3rd place winner** of State Farm's Sponsor Challenge at Shell Hacks 2025 out of **50+** projects submitted.
- Created an AI voice agent delivering interactive, personalized feedback across **3** different insurance categories.
- Implemented backend services to process and structure conversational data generating dynamic agent responses, reducing response time by **66%** using **Python**, integrated with the **Vapi API**.

TECHNICAL SKILLS

Languages: C, C++, Assembly, Python, Java, C#, JavaScript, TypeScript, HTML, CSS

GPU & Systems: HIP, CUDA, ROCm, rocprof, OpenGL, CMake

Frameworks: Unreal Engine, Unity, Next.js, Flask, Node.js, OpenCV, Mediapipe, Three.js

Libraries & Tools: React, Tailwind CSS, SDL2, WebSockets, TFTP.eSPI, Arduino, Docker, AWS, Git, Linux, ESP32, PlatformIO, Cloudflare, Vite, Photon RPC